

DP Computer Science: A 1.1.5 Fetch, Decode, Execute Cycle

1. Core Concepts & Learning Objectives

Learning Intentions

By the end of this lesson, students will be able to:

- 1. Describe the **three stages** of the cycle in the correct order
- 2. Name the registers (**PC, MAR, MDR, IR, and the accumulator**) and state what each holds at the point it acts
- 3. Describe the **job and direction** of the address, data, and control buses during the fetch and execute stages
- 4. Give a **detailed account** of one instruction passing through the full cycle from fetch to result

Command Term Note: The lesson is held at **Describe**—a detailed, ordered account. Pipelining is moved to A1.1.6, as anything beyond a detailed account reaches past this statement.

Key Vocabulary (Tier 3)

Term	Definition
Register	A small, very fast store inside the CPU that holds one value at a time
Program Counter (PC)	Holds the memory address of the next instruction to fetch
Memory Address Register (MAR)	Holds the address of the memory location about to be used
Memory Data Register (MDR)	Holds the instruction or data just read from memory, or about to be written to it
Instruction Register (IR)	Holds the current instruction while it is decoded and executed

Term	Definition
Opcode and Operand	The two parts of a machine instruction; the opcode is the operation, the operand is the data or address it acts on

Pedagogical Focus

- **Communication (ATL):** Students consciously practise communicating a technical process in the right order with the right terms. Name it at the start: *"Today you are practising giving a precise, ordered account, first spoken, then written."*
- **Inquiry-Based Learning (ATT):** Students meet a genuine question and investigate it on the simulator before any formal explanation. The order is **explore then explain**. The investigation is guided with fixed checkpoints rather than left open, because novices need a structured route, not free discovery.

Secondary ATLs present: Thinking skills (sequencing the trace) and self-management (working through simulator checkpoints).

EAL Scaffold

Support Type	Description
Sentence Frame	"During the fetch stage, the address in the ___ is copied to the ___. The address travels along the ___ bus while a ___ signal is sent on the ___ bus. The instruction returns along the ___ bus into the ___, is copied into the ___, and the ___ is incremented."
Idiom Flag	Avoid "the heartbeat of the processor" and "the CPU's dance." Use plain alternative: "the repeating cycle the CPU runs for every instruction." Avoid "Explain it like I'm 5"—use "Describe each stage using the correct register and bus names."

2. Lesson Sequence (One-Hour Block)

Part 1: Do Now (Retrieval) — ~10%

Activity: Three questions recalling prior learning:

Question	Source
1. Name the two main units inside the CPU and state what each does	[A1.1.1]
2. What is a register, and why is it faster to use than main memory?	[A1.1.1]
3. Name the three buses that connect the CPU and main memory	[A1.1.1]

Note: Assumes the running order on the unit overview, where A1.1.1 precedes this lesson.

Part 2: Main Sequence — ~60%

Phase A: Pose the Puzzle (~5 mins)

Question to students:

"A program is only numbers sitting in memory. How does the CPU turn those numbers into actions, one instruction at a time, without ever losing the order?"

Activity: Students predict before running anything on the simulator. This creates a genuine cognitive hook.

Phase B: Structured Exploration with Checkpoints (~15 mins)

Tool: Custom CPU Simulator (built to tie in directly with the syllabus)

Activity: Pairs step the program through one instruction at a time and complete a capture sheet.

Checkpoint	Focus Questions
A — After the Fetch	Which register supplied the address? Where did the instruction end up? What changed in the PC?

Checkpoint

Focus Questions

B — After the Decode

What two parts did the instruction split into?

C — After the Execute

Where did the result go?

Teacher Role: Circulate and confirm each checkpoint before pairs move on.

Note: The program is pre-loaded and the capture sheet is held to three checkpoints to keep the investigation inside the time.

Phase C: Explanation That Consolidates (~15 mins)

Using the deck and eBook stage descriptions:

1. The class narrates the cycle back while you fix the precise names
2. **Address the MAR and MDR swap directly** using the students' own traces as the evidence

The Complete Cycle:

text

FETCH STAGE

1. PC → MAR (address of next instruction)
2. Address travels on address bus
3. Read signal sent on control bus
4. Instruction returns on data bus → MDR
5. Instruction copied to IR
6. PC incremented



DECODE STAGE

1. Instruction in IR is split:
 - Opcode (operation to perform)
 - Operand (data or address to act on)
2. Control Unit generates control signals

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EXECUTE STAGE

1. Values routed through ALU
2. ALU performs the operation
3. Result stored in accumulator or memory

Part 3: Deliberate Practice — ~20%

Activity A — Cycle Sequencing Worksheet:

- Students rearrange jumbled operations into the correct order
- Write a brief explanation of what happens in each stage

Activity B — Written Description:

Students write a short description of the **fetch stage** using the sentence frame:

"During the fetch stage, the address in the ___ is copied to the ___. The address travels along the ___ bus while a ___ signal is sent on the ___ bus. The instruction returns along the ___ bus into the ___, is copied into the ___, and the ___ is incremented."

Protect this phase. It is where the ordered account consolidates.

Part 4: Exit Check — ~10%

One Markable Question:

"Describe the steps the CPU performs during the fetch stage."

Alternative: eBook TRY IT ON PAPER question 2 (4-mark version with mark points).

Teacher Note: Read responses the same day. An answer that sends the instruction to the MAR or the address to the MDR tells you to reopen the register roles next lesson.

3. Summary: The Three Stages

Stage	Key Actions	Registers/Buses Involved
Fetch	PC → MAR → address bus; instruction returns via data bus → MDR → IR; PC incremented	PC, MAR, MDR, IR; address bus, data bus, control bus (read signal)
Decode	Instruction split into opcode and operand; CU generates control signals	IR, CU
Execute	Values routed through ALU; operation performed; result stored	ALU, accumulator, MDR (for memory writes)

4. Differentiation & Extension

Support (Scaffolding That Can Be Removed)

Support Strategy	Description
Part-Completed Capture Sheet	Some fields pre-filled
Sentence Frame	For the written description
Name Bank	List of registers and buses to choose from

Withdraw the frame once pairs are tracing fluently, so the support is temporary, not a permanent simpler version.

Challenge (Deeper at the Same Command Term)

Challenge Strategy	Description
Trace a Memory Write	Trace an instruction that writes its result back to memory; describe the data bus carrying in the other direction and

Challenge Strategy	Description
	all three buses acting in the execute stage
Trace Two Instructions	Describe how the incremented PC carries the order across multiple instructions

Keep this at Describe and away from pipelining (A1.1.6, HL) and the CPU-vs-GPU comparison (A1.1.3, HL).

5. Engagement Activities (Supplementary)

If time allows, these can replace or supplement the main sequence:

Activity	Description
"Instruction Charades"	Students act out simple instructions, simulating CPU actions during fetch, decode, and execute
"The Repeating Cycle" Activity	Use an analogy of a simple choreography to illustrate a repetitive cycle with distinct stages
Interactive Slideshow Exploration	Present diagrams, animations, and examples
"CPU Simulator" Exploration	Step through execution of simple instructions and observe data flow

Simulator Options:

- Custom CPU Simulator (recommended — ties in directly with syllabus)
- Little Man Computer (LMC)

6. Learning Objectives (Summary)

Knowledge and Understanding

- Describe the fetch-decode-execute cycle as the fundamental process by which CPUs execute instructions
- Explain the three stages and the operations performed in each
- Identify the roles of key CPU components (CU, ALU, registers, buses) in the cycle
- Explain how memory and registers interact via the address, data, and control buses

Skills

- Analyse the data flow and instructions within the CPU during the cycle
- Interpret simple machine language instructions and explain how they are processed
- Trace the execution of instructions through the different stages of the cycle

ATL Skills

Skill	Description
Communication	Effectively communicate understanding through discussions and written explanations
Collaboration	Work collaboratively to analyse diagrams, solve problems, and complete activities
Critical Thinking	Analyse the sequence of operations; evaluate importance in computer processing
Information Literacy	Use online resources and research to deepen understanding

7. Resources

- Custom CPU Simulator (tied directly to syllabus)
- Presentation (Slide Deck)
- InThinking eBook (A 1.1.5 with quiz)
- Worksheet: Cycle Sequencing (with answers)
- Quizlet: Vocabulary flashcards

Author's Reflection

"I have built this lesson around the custom CPU Simulator rather than the slide deck and the Little Man Computer (LMC), because the cycle is one of the few Describe statements where students can watch the mechanism work before anyone names it. The LMC did not follow the abstracted form of the CPU from the syllabus and was a little esoteric and confusing for some students.

The investigation runs first and the slides consolidate afterwards, which is the order inquiry calls for. I no longer present the content up front and instead use the deck to fix the precise terms once students have something to attach them to.

The genuine trade-off is time. Stepping through the CPU Simulator in a 45-minute block is slower than talking through the five fetch steps, so the program is pre-loaded and the capture sheet is held to three checkpoints to keep the investigation inside the time. Students meet the MAR and the MDR as they have seen in action, rather than more acronyms to memorise.

I have also held the whole lesson at Describe and moved pipelining (which was included previously) to A1.1.6, because anything beyond a detailed, ordered account, pipelining included, reaches past this statement and returns nothing in the exam."